Раздел 1

1. Создание ключа ssh:

```
eltex-pg2-v2@eltex-2026-autumn:~$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/eltex-pg2-v2/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/eltex-pg2-v2/.ssh/id_rsa
Your public key has been saved in /home/eltex-pg2-v2/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:tcLQoxoTzrTSu0adQieMkaIm/bjdy9pJfsRQgM/nKWM eltex-pg2-v2@eltex-2026-au
tumn
The key's randomart image is:
+---[RSA 3072]----+
|    . ...        |
|. o .  ..        |
|.o +oo..o .      |
|o.o=+o=+.o .     |
|o .+B+.*S..      |
|  ..+=E =.       |
|   +o+.+         |
|  . o*...        |
|   .o.*o         |
+----[SHA256]-----+
eltex-pg2-v2@eltex-2026-autumn:~$ ls -la ~/.ssh/
total 20
drwx------  2 eltex-pg2-v2 eltex-pg2-v2 4096 Oct 25 22:06 .
drwx------ 17 eltex-pg2-v2 eltex-pg2-v2 4096 Oct 19 22:08 ..
-rw-------  1 eltex-pg2-v2 eltex-pg2-v2  412 Oct 14 14:56 authorized_keys
-rw-------  1 eltex-pg2-v2 eltex-pg2-v2 2622 Oct 25 22:06 id_rsa
-rw-r--r--  1 eltex-pg2-v2 eltex-pg2-v2  584 Oct 25 22:06 id_rsa.pub
```

Рисунок 1 – Создание SSH-ключа на сервере 1 с помощью утилиты ssh-keygen

2. Копирование созданного ключа на сервер для 2-й практики:

```
eltex-pg2-v2@eltex-2026-autumn:~$ ssh-copy-id -i .ssh/id_rsa.pub root@172.16.
9.202
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ".ssh/id_rsa.pu
b"
The authenticity of host '172.16.9.202 (172.16.9.202)' can't be established.
ED25519 key fingerprint is SHA256:YNVzj+RtetZfXv5XUn/CUjeEcaMD4eg3jy8EuGGIS+o
.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filt
er out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are pro
mpted now it is to install the new keys
root@172.16.9.202's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'root@172.16.9.202'"
and check to make sure that only the key(s) you wanted were added.

eltex-pg2-v2@eltex-2026-autumn:~$ ssh root@172.16.9.202 'ls -la ~/.ssh/author
ized_keys; cat ~/.ssh/authorized_keys'
-rw------- 1 root root 584 Oct 25 15:13 /root/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQC+jhMhwuwDy8ORVtK4Rxtvo55IoZuHEZhTsxgm8
Gwoumhr8PnuBuChkTmzVzo/GUBaytUDkb78N3RHocnHCEp3tuD3MV/zYMM1NHiqeAaXeroGN0IWYg
m4PMagA3669EGjWg4rJN4iXNAKdJkH9qsLHcaJX+INAgS7PekUz2JcBJpfbooB8SFug88h5e/ZgnR
t3lBDF4QoVesolX29laLFy/SEnelXZgiBIL4tiE8bMc9+CCXu33B8gjiuyguuwreb8HD7AtjnEzFO
ERazpZZJcDVUUIGvWRSDS4tIdChI0VsKn5bl5U3hSDC745lemxGpc9j5i5VUd+GRQ/o9QsaPml+le
6mCdAmV2NDY2KHdbqkmMsNRaSnandUjKkvWWI0cNLTQ/EqACK2STNCz6DnIbzQ++rK1QUHILIHR+3
9b9D7oxkgHgV3g9Hvoaqk+ZocoJMx43Hip2Vtk8IQKhnFOlThZPnKCrF3vduj0KqBhLTpsSzCGyxs
i5vzOL1IMHBk= eltex-pg2-v2@eltex-2026-autumn
```

Рисунок 2 – Копирование публичного ключа на сервер 2 (практика 2) с помощью ssh-copy-id

3. Сравнение содержимого файла открытого ключа на сервере 1 и файла authorized_keys на сервере для 2-й практики, а также прав доступа для каждого из файлов:

```
eltex-pg2-v2@eltex-2026-autumn:~$ ssh root@172.16.9.202 'ls -la ~/.ssh/author
ized_keys; cat ~/.ssh/authorized_keys'
-rw------- 1 root root 584 Oct 25 15:13 /root/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQC+jhMhwuwDy8ORVtK4Rxtvo55IoZuHEZhTsxgm8
Gwoumhr8PnuBuChkTmzVzo/GUBaytUDkb78N3RHocnHCEp3tuD3MV/zYMM1NHiqeAaXeroGN0IWYg
m4PMagA3669EGjWg4rJN4iXNAKdJkH9qsLHcaJX+INAgS7PekUz2JcBJpfbooB8SFug88h5e/ZgnR
t3lBDF4QoVesolX29laLFy/SEnelXZgiBIL4tiE8bMc9+CCXu33B8gjiuyguuwreb8HD7AtjnEzFO
ERazpZZJcDVUUIGvWRSDS4tIdChI0VsKn5bl5U3hSDC745lemxGpc9j5i5VUd+GRQ/o9QsaPml+le
6mCdAmV2NDY2KHdbqkmMsNRaSnandUjKkvWWI0cNLTQ/EqACK2STNCz6DnIbzQ++rK1QUHILIHR+3
9b9D7oxkgHgV3g9Hvoaqk+ZocoJMx43Hip2Vtk8IQKhnFOlThZPnKCrF3vduj0KqBhLTpsSzCGyxs
i5vzOL1IMHBk= eltex-pg2-v2@eltex-2026-autumn
eltex-pg2-v2@eltex-2026-autumn:~$ ls -la .ssh/id_rsa.pub
-rw-r--r-- 1 eltex-pg2-v2 eltex-pg2-v2 584 Oct 25 22:06 .ssh/id_rsa.pub
eltex-pg2-v2@eltex-2026-autumn:~$ cat .ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQC+jhMhwuwDy8ORVtK4Rxtvo55IoZuHEZhTsxgm8
Gwoumhr8PnuBuChkTmzVzo/GUBaytUDkb78N3RHocnHCEp3tuD3MV/zYMM1NHiqeAaXeroGN0IWYg
m4PMagA3669EGjWg4rJN4iXNAKdJkH9qsLHcaJX+INAgS7PekUz2JcBJpfbooB8SFug88h5e/ZgnR
t3lBDF4QoVesolX29laLFy/SEnelXZgiBIL4tiE8bMc9+CCXu33B8gjiuyguuwreb8HD7AtjnEzFO
ERazpZZJcDVUUIGvWRSDS4tIdChI0VsKn5bl5U3hSDC745lemxGpc9j5i5VUd+GRQ/o9QsaPml+le
6mCdAmV2NDY2KHdbqkmMsNRaSnandUjKkvWWI0cNLTQ/EqACK2STNCz6DnIbzQ++rK1QUHILIHR+3
9b9D7oxkgHgV3g9Hvoaqk+ZocoJMx43Hip2Vtk8IQKhnFOlThZPnKCrF3vduj0KqBhLTpsSzCGyxs
i5vzOL1IMHBk= eltex-pg2-v2@eltex-2026-autumn
```

Рисунок 3 – Сравнение содержимого публичного ключа на сервере 1 (id_rsa.pub) и файла authorized_keys на сервере 2, а также проверка прав доступа

4. Создание пользователя user1:

```
root@eltex-practice2-pg2-v2:~# useradd -m -s /bin/bash user1
root@eltex-practice2-pg2-v2:~# passwd user1
New password:
Retype new password:
passwd: password updated successfully
root@eltex-practice2-pg2-v2:~# id user1
uid=1001(user1) gid=1001(user1) groups=1001(user1)
root@eltex-practice2-pg2-v2:~# getent passwd user1
user1:x:1001:1001::/home/user1:/bin/bash
root@eltex-practice2-pg2-v2:~# ls -ld /home/user1
drwxr-x--- 2 user1 user1 4096 Oct 25 15:26 /home/user1
```

Рисунок 4 – Создание пользователя user1 командой useradd с домашним каталогом и оболочкой /bin/bash, установка пароля и проверка

5. Создание пользователей user2 и user3:

```
root@eltex-practice2-pg2-v2:~# adduser user2
info: Adding user `user2' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user2' (1002) ...
info: Adding new user `user2' (1002) with group `user2 (1002)' ...
info: Creating home directory `/home/user2' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user2
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `user2' to supplemental / extra groups `users' ...
info: Adding user `user2' to group `users' ...
```

Рисунок 5 – Создание пользователя user2 с помощью интерактивной команды adduser

```
root@eltex-practice2-pg2-v2:~# adduser user3
info: Adding user `user3' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user3' (1003) ...
info: Adding new user `user3' (1003) with group `user3 (1003)' ...
info: Creating home directory `/home/user3' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user3
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `user3' to supplemental / extra groups `users' ...
info: Adding user `user3' to group `users' ...
```

Рисунок 6 – Создание пользователя user3 с помощью команды adduser

```
root@eltex-practice2-pg2-v2:~# id user2
uid=1002(user2) gid=1002(user2) groups=1002(user2),100(users)
root@eltex-practice2-pg2-v2:~# id user3
uid=1003(user3) gid=1003(user3) groups=1003(user3),100(users)
root@eltex-practice2-pg2-v2:~# getent passwd user2 user3
user2:x:1002:1002:,,,:/home/user2:/bin/bash
user3:x:1003:1003:,,,:/home/user3:/bin/bash
root@eltex-practice2-pg2-v2:~# ls -ld /home/user2 /home/user3
drwxr-x--- 2 user2 user2 4096 Oct 25 15:33 /home/user2
drwxr-x--- 2 user3 user3 4096 Oct 25 15:34 /home/user3
```

Рисунок 7 – Проверка параметров созданных пользователей user2 и user3 (id, getent, ls -ld)

6. Смена shell на nologin для user3, вход под этим пользователем, logout:

```
root@eltex-practice2-pg2-v2:~# usermod -s /usr/sbin/nologin user3
root@eltex-practice2-pg2-v2:~# getent passwd user3
user3:x:1003:1003:,,,:/home/user3:/usr/sbin/nologin
root@eltex-practice2-pg2-v2:~# su - user3
This account is currently not available.
root@eltex-practice2-pg2-v2:~# su -s /bin/bash user3
user3@eltex-practice2-pg2-v2:/root$ whoami
user3
user3@eltex-practice2-pg2-v2:/root$ echo $SHELL
/bin/bash
user3@eltex-practice2-pg2-v2:/root$ exit
exit
```

Рисунок 8 – Смена оболочки для user3 на /usr/sbin/nologin и проверка входа под ним с помощью su и su -s /bin/bash

7. Создание новой группы, добавление в неё всех пользователей user*, просмотр список групп всех пользователей user*:

```
root@eltex-practice2-pg2-v2:~# groupadd sharegrp
root@eltex-practice2-pg2-v2:~# usermod -aG sharegrp user1
root@eltex-practice2-pg2-v2:~# usermod -aG sharegrp user2
^[[Aroot@eltex-practice2-pg2-v2:~# usermod -aG sharegrp user3
root@eltex-practice2-pg2-v2:~# groups user1 user2 user3
user1 : user1 sharegrp
user2 : user2 users sharegrp
user3 : user3 users sharegrp
root@eltex-practice2-pg2-v2:~# id user1
uid=1001(user1) gid=1001(user1) groups=1001(user1),1004(sharegrp)
root@eltex-practice2-pg2-v2:~# id user2
uid=1002(user2) gid=1002(user2) groups=1002(user2),100(users),1004(sharegrp)
root@eltex-practice2-pg2-v2:~# id user3
uid=1003(user3) gid=1003(user3) groups=1003(user3),100(users),1004(sharegrp)
root@eltex-practice2-pg2-v2:~# getent group sharegrp
sharegrp:x:1004:user1,user2,user3
```

Рисунок 9 – Создание группы sharegrp, добавление её всем пользователям user как дополнительной и проверка списка групп*

9. Создание каталога /opt/share, назначение группы из предыдущего пункта владельцем, установка на этот каталог бита SGID, прав доступа rwx:

```
root@eltex-practice2-pg2-v2:~# mkdir -p /opt/share
root@eltex-practice2-pg2-v2:~# chgrp sharegrp /opt/share
root@eltex-practice2-pg2-v2:~# chmod 2770 /opt/share
root@eltex-practice2-pg2-v2:~# ls -ld /opt/share
drwxrws--- 2 root sharegrp 4096 Oct 25 16:15 /opt/share
root@eltex-practice2-pg2-v2:~# stat -c '%A %U:%G %n' /opt/share
drwxrws--- root:sharegrp /opt/share
root@eltex-practice2-pg2-v2:~# getent group sharegrp
sharegrp:x:1004:user1,user2,user3
```

Рисунок 10 – Создание каталога /opt/share, назначение группы sharegrp владельцем и установка бита SGID с правами rwx (chmod 2770)

10. Задание для user1 umask:

```
root@eltex-practice2-pg2-v2:~# grep -n 'umask' /home/user1/.bashrc
118:umask 027
root@eltex-practice2-pg2-v2:~# su - user1
user1@eltex-practice2-pg2-v2:~$ umask
0027
user1@eltex-practice2-pg2-v2:~$ touch /home/user1/test_umask_clean
user1@eltex-practice2-pg2-v2:~$ touch /home/user1/test_umask
user1@eltex-practice2-pg2-v2:~$ ls -l /home/user1/test_umask
-rw-r----- 1 user1 user1 0 Oct 25 17:12 /home/user1/test_umask
user1@eltex-practice2-pg2-v2:~$ exit
logout
```

Рисунок 11 – Настройка перманентного umask 027 для user1 в ~/.bashrc и проверка прав создаваемого файла

11. Создание новых файлов в каталоге /opt/share каждый пользователем, удаление файлов, созданных другими пользователями:

```
root@eltex-practice2-pg2-v2:~# su - user1
user1@eltex-practice2-pg2-v2:~$ cd /opt/share && pwd && touch file_user1.txt
&& ls -l
/opt/share
total 0
-rw-r----- 1 user1 sharegrp 0 Oct 25 17:23 file_user1.txt
user1@eltex-practice2-pg2-v2:/opt/share$ exit
logout
root@eltex-practice2-pg2-v2:~# su - user2
user2@eltex-practice2-pg2-v2:~$ cd /opt/share && touch file_user2.txt && ls -
l
total 0
-rw-r----- 1 user1 sharegrp 0 Oct 25 17:23 file_user1.txt
-rw-rw-r-- 1 user2 sharegrp 0 Oct 25 17:24 file_user2.txt
user2@eltex-practice2-pg2-v2:/opt/share$ exit
logout
root@eltex-practice2-pg2-v2:~# su -s /bin/bash user3
user3@eltex-practice2-pg2-v2:/root$ cd /opt/share && touch file_user3.txt &&
ls -l
total 0
-rw-r----- 1 user1 sharegrp 0 Oct 25 17:23 file_user1.txt
-rw-rw-r-- 1 user2 sharegrp 0 Oct 25 17:24 file_user2.txt
-rw-rw-r-- 1 user3 sharegrp 0 Oct 25 17:24 file_user3.txt
user3@eltex-practice2-pg2-v2:/opt/share$ exit
exit
root@eltex-practice2-pg2-v2:~# su - user1
user1@eltex-practice2-pg2-v2:~$ cd /opt/share
user1@eltex-practice2-pg2-v2:/opt/share$ rm -f file_user2.txt
user1@eltex-practice2-pg2-v2:/opt/share$ rm -f file_user3.txt
user1@eltex-practice2-pg2-v2:/opt/share$ ls -l
total 0
-rw-r----- 1 user1 sharegrp 0 Oct 25 17:23 file_user1.txt
user1@eltex-practice2-pg2-v2:/opt/share$ exit
logout
```

Рисунок 12 – Создание каждым из пользователей (user1, user2, user3) своих файлов в каталоге /opt/share

12. Повторение предыдущего пункта с предварительно установленным sticky bit на каталоге /opt/share:

```
root@eltex-practice2-pg2-v2:~# su - user1
user1@eltex-practice2-pg2-v2:~$ cd /opt/share
user1@eltex-practice2-pg2-v2:/opt/share$ rm -f file_user2.txt
user1@eltex-practice2-pg2-v2:/opt/share$ rm -f file_user3.txt
user1@eltex-practice2-pg2-v2:/opt/share$ ls -l
total 0
-rw-r----- 1 user1 sharegrp 0 Oct 25 17:23 file_user1.txt
user1@eltex-practice2-pg2-v2:/opt/share$ exit
logout
root@eltex-practice2-pg2-v2:~# su - user2
user2@eltex-practice2-pg2-v2:~$ cd /opt/share && touch file_user2.txt
user2@eltex-practice2-pg2-v2:/opt/share$ exit
logout
root@eltex-practice2-pg2-v2:~# su -s /bin/bash user3
user3@eltex-practice2-pg2-v2:/root$ cd /opt/share && touch file_user3.txt
user3@eltex-practice2-pg2-v2:/opt/share$ exit
exit
root@eltex-practice2-pg2-v2:~# chmod 3770 /opt/share
root@eltex-practice2-pg2-v2:~# ls -ld /opt/share
drwxrws--T 2 root sharegrp 4096 Oct 25 17:32 /opt/share
root@eltex-practice2-pg2-v2:~# stat -c '%A %a %U:%G %n' /opt/share
drwxrws--T 3770 root:sharegrp /opt/share
root@eltex-practice2-pg2-v2:~# su - user1
user1@eltex-practice2-pg2-v2:~$ cd /opt/share
user1@eltex-practice2-pg2-v2:/opt/share$ rm -f file_user2.txt file_user3.txt
rm: cannot remove 'file_user2.txt': Operation not permitted
rm: cannot remove 'file_user3.txt': Operation not permitted
user1@eltex-practice2-pg2-v2:/opt/share$ ls -ls
total 0
0 -rw-r----- 1 user1 sharegrp 0 Oct 25 17:23 file_user1.txt
0 -rw-rw-r-- 1 user2 sharegrp 0 Oct 25 17:30 file_user2.txt
0 -rw-rw-r-- 1 user3 sharegrp 0 Oct 25 17:32 file_user3.txt
user1@eltex-practice2-pg2-v2:/opt/share$ ls -l
total 0
-rw-r----- 1 user1 sharegrp 0 Oct 25 17:23 file_user1.txt
-rw-rw-r-- 1 user2 sharegrp 0 Oct 25 17:30 file_user2.txt
-rw-rw-r-- 1 user3 sharegrp 0 Oct 25 17:32 file_user3.txt
user1@eltex-practice2-pg2-v2:/opt/share$ exit
logout
```

Рисунок 13 – Попытка удаления файлов, созданных другими пользователями, до установки sticky bit (удаление проходит успешно)

13. Разрешение для user1 и user2 выполнять привилегированную команду dmesg:

```
root@eltex-practice2-pg2-v2:~# which sudo dmesg
/usr/bin/sudo
/usr/bin/dmesg
root@eltex-practice2-pg2-v2:~# cat /proc/sys/kernel/dmesg_restrict
1
```

Рисунок 14 – Проверка наличия sudo и dmesg, а также параметра dmesg_restrict

```
root@eltex-practice2-pg2-v2:~# echo 'user1 ALL=(ALL) NOPASSWD: /usr/bin/dmesg
' > /etc/sudoers.d/user1_dmesg
root@eltex-practice2-pg2-v2:~# chmod 440 /etc/sudoers.d/user1_dmesg
root@eltex-practice2-pg2-v2:~# visudo -c
s/etc/sudoers: parsed OK
/etc/sudoers.d/README: parsed OK
/etc/sudoers.d/user1_dmesg: parsed OK
root@eltex-practice2-pg2-v2:~# su - user1
user1@eltex-practice2-pg2-v2:~$ sudo -l
Matching Defaults entries for user1 on eltex-practice2-pg2-v2:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:
/bin\:/snap/bin,
    use_pty

User user1 may run the following commands on eltex-practice2-pg2-v2:
    (ALL) NOPASSWD: /usr/bin/dmesg
user1@eltex-practice2-pg2-v2:~$ sudo dmeg | head -5
[sudo] password for user1:
sudo: dmeg: command not found
user1@eltex-practice2-pg2-v2:~$ sudo dmesg | head -5
[    0.000000] Linux version 6.8.0-147-generic (buildd@lcy02-amd64-081) (x86_
64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0, GNU ld (GNU Binu
tils for Ubuntu) 2.42) #147-Ubuntu SMP PREEMPT_DYNAMIC Thu Sep 24 11:33:16 UT
C 2026 (Ubuntu 6.8.0-147.147-generic 6.8.12)
[    0.000000] Command line: BOOT_IMAGE=/vmlinuz-6.8.0-147-generic root=/dev/
mapper/ubuntu--vg-ubuntu--lv ro
[    0.000000] KERNEL supported cpus:
[    0.000000]   Intel GenuineIntel
[    0.000000]   AMD AuthenticAMD
user1@eltex-practice2-pg2-v2:~$ exit
logout
```

Рисунок 15 – Разрешение user1 выполнять dmesg через sudo (создание файла в /etc/sudoers.d/), проверка синтаксиса visudo -c и тестовый запуск (включая случайную опечатку dmeg)

```
root@eltex-practice2-pg2-v2:~# cat > /usr/local/bin/dmesg_user2.sh << 'EOF'
> #!/bin/bash
> exec /usr/bin/dmesg
> EOF
root@eltex-practice2-pg2-v2:~# chmod 4755 /usr/local/bin/dmesg_user2.sh
root@eltex-practice2-pg2-v2:~# ls -l /usr/local/bin/dmesg_user2.sh
-rwsr-xr-x 1 root root 32 Oct 25 17:46 /usr/local/bin/dmesg_user2.sh
root@eltex-practice2-pg2-v2:~# su - user2
user2@eltex-practice2-pg2-v2:~$ id
uid=1002(user2) gid=1002(user2) groups=1002(user2),100(users),1004(sharegrp)
user2@eltex-practice2-pg2-v2:~$ /usr/local/bin/dmesg_user2.sh | head -5
dmesg: read kernel buffer failed: Operation not permitted
user2@eltex-practice2-pg2-v2:~$ exit
logout
```

Рисунок 16 – Создание bash-скрипта с SUID-битом для user2 и проверка его работы (ядро игнорирует SUID на скриптах)

14. Задание для всех пользователей user* время действия пароля:

```
root@eltex-practice2-pg2-v2:~# chage -M 10 user1
root@eltex-practice2-pg2-v2:~# chage -M 10 user2
root@eltex-practice2-pg2-v2:~# chage -M 10 user3
root@eltex-practice2-pg2-v2:~# chage -l user1
Last password change                                    : Oct 25, 2026
Password expires                                        : Nov 04, 2026
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 10
Number of days of warning before password expires       : 7
root@eltex-practice2-pg2-v2:~# chage -l user2
Last password change                                    : Oct 25, 2026
Password expires                                        : Nov 04, 2026
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 10
Number of days of warning before password expires       : 7
root@eltex-practice2-pg2-v2:~# chage -l user3
Last password change                                    : Oct 25, 2026
Password expires                                        : Nov 04, 2026
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 10
Number of days of warning before password expires       : 7
root@eltex-practice2-pg2-v2:~# grep -E '^user[123]:' /etc/shadow | awk -F: '{
print $1": max="$5}'
user1: max=10
user2: max=10
user3: max=10
```

Рисунок 17 – Установка срока действия пароля 10 дней для всех пользователей user (chage -M 10) и проверка изменений*

15. Редактирование файла /etc/motd:

```
root@eltex-practice2-pg2-v2:~# cat > /etc/motd << 'EOF'
> Практика 2
> Конинский Алексей
> EOF
root@eltex-practice2-pg2-v2:~# cat /etc/motd
Практика 2
Конинский Алексей
```

Рисунок 18 – Редактирование файла /etc/motd (вписаны номер практики и ФИО)

16. Создание копии содержимого каталога /etc в каталог /root/etc_backup при помощи rsync:

```
root@eltex-practice2-pg2-v2:~# which rsync || apt install -y rsync
/usr/bin/rsync
root@eltex-practice2-pg2-v2:~# mkdir -p /root/etc_backup
root@eltex-practice2-pg2-v2:~# rsync -a /etc/ /root/etc_backup/
root@eltex-practice2-pg2-v2:~# ls -ld /root/etc_backup
drwxr-xr-x 109 root root 4096 Oct 25 18:22 /root/etc_backup
root@eltex-practice2-pg2-v2:~# ls /root/etc_backup | head
adduser.conf
alternatives
apparmor
apparmor.d
apport
apt
bash.bashrc
bash_completion
bash_completion.d
bindresvport.blacklist
root@eltex-practice2-pg2-v2:~# du -sh /etc/ root/etc_backup
6.1M    /etc/
du: cannot access 'root/etc_backup': No such file or directory
root@eltex-practice2-pg2-v2:~# diff -q /etc/passwd /root/etc_backup/passwd &&
 echo "passwd OK"
passwd OK
root@eltex-practice2-pg2-v2:~# diff -q /etc/motd /root/etc_backup/motd && ech
o "motd OK"
motd OK
```

Рисунок 19 – Копирование содержимого каталога /etc в /root/etc_backup с помощью rsync и проверка diff

17. Архивирование содержимого каталога /root/etc_backup архиватором tar алгоритмами gzip, bzip2, 7zip и сравнение размеров полученных файлов:

```
root@eltex-practice2-pg2-v2:~# which tar gzip bzip2 7z
/usr/bin/tar
/usr/bin/gzip
/usr/bin/bzip2
/usr/bin/7z
root@eltex-practice2-pg2-v2:~# rm -f etc_backup.tar.gz etc_backup.tar.bz2 etc
_backup.tar.7z
root@eltex-practice2-pg2-v2:~# tar -czf etc_backup.tar.gz  etc_backup
root@eltex-practice2-pg2-v2:~# tar -cjf etc_backup.tar.bz2 etc_backup
root@eltex-practice2-pg2-v2:~# 7z a -snl -t7z etc_backup.tar.7z etc_backup >
/dev/null
root@eltex-practice2-pg2-v2:~# echo "7z exit=$?"
7z exit=0
root@eltex-practice2-pg2-v2:~# ls -lh etc_backup.tar.gz etc_backup.tar.bz2 et
c_backup.tar.7z
-rw-r--r-- 1 root root 439K Oct 25 19:00 etc_backup.tar.7z
-rw-r--r-- 1 root root 526K Oct 25 19:00 etc_backup.tar.bz2
-rw-r--r-- 1 root root 563K Oct 25 19:00 etc_backup.tar.gz
root@eltex-practice2-pg2-v2:~# du -sh etc_backup
6.1M    etc_backup
root@eltex-practice2-pg2-v2:~# for f in etc_backup.tar.gz etc_backup.tar.bz2
etc_backup.tar.7z; do
> printf '%-25s %10s bytes\n' "$f" "$(stat -c%s "$f")"
> done
etc_backup.tar.gz            576436 bytes
etc_backup.tar.bz2           538456 bytes
etc_backup.tar.7z            448992 bytes
root@eltex-practice2-pg2-v2:~# tar -tzf etc_backup.tar.gz  | head -3
etc_backup/
etc_backup/rc4.d/
etc_backup/rc4.d/S01uuidd
root@eltex-practice2-pg2-v2:~# tar -tjf etc_backup.tar.bz2 | head -3
etc_backup/
etc_backup/rc4.d/
etc_backup/rc4.d/S01uuidd
root@eltex-practice2-pg2-v2:~# 7z l etc_backup.tar.7z | grep -E 'passwd|motd'
 | head -5
2026-10-15 09:35:57 D....            0            0  etc_backup/update-motd.d
2025-02-16 20:51:18 ....A            0            0  etc_backup/security/opas
swd
2025-02-16 21:04:48 ....A          682               etc_backup/default/motd-
news
2026-10-25 18:22:52 ....A           53               etc_backup/motd
2024-02-22 12:30:23 ....A           92               etc_backup/pam.d/chpassw
d
```

Рисунок 20 – Архивирование каталога /root/etc_backup с помощью tar (gzip, bzip2) и 7z, сравнение размеров архивов

18. Редактирование файла /etc/motd, синхронизирование каталога /root/etc_backup с каталогом /etc при помощи rsync, добавление файла motd в архив, сжатый gzip:

```
root@eltex-practice2-pg2-v2:~# echo "Обновлено $(date '+%Y-%m-%d %H:%M:%S')"
>> /etc/motd
root@eltex-practice2-pg2-v2:~# cat /etc/motd
Практика 2
Конинский Алексей
Обновлено 2026-10-25 20:27:50
root@eltex-practice2-pg2-v2:~# rsync -a /etc/ /root/etc_backup/
root@eltex-practice2-pg2-v2:~# diff -q /etc/motd /root/etc_backup/motd && ech
o "motd synced"
motd synced
root@eltex-practice2-pg2-v2:~# ls -l /etc/motd /root/etc_backup/motd
-rw-r--r-- 1 root root 92 Oct 25 20:27 /etc/motd
-rw-r--r-- 1 root root 92 Oct 25 20:27 /root/etc_backup/motd
root@eltex-practice2-pg2-v2:~# rm -f etc_backup.tar.gz
root@eltex-practice2-pg2-v2:~# tar -czf etc_backup.tar.gz etc_backup
root@eltex-practice2-pg2-v2:~# tar -tzf etc_backup.tar.gz | grep 'etc_backup/
motd$'
etc_backup/motd
root@eltex-practice2-pg2-v2:~# tar -xzf etc_backup.tar.gz -O etc_backup/motd
Практика 2
Конинский Алексей
Обновлено 2026-10-25 20:27:50
root@eltex-practice2-pg2-v2:~# ls -lh etc_backup.tar.gz
-rw-r--r-- 1 root root 563K Oct 25 20:29 etc_backup.tar.gz
```

Рисунок 21 – Добавление текущей даты/времени в /etc/motd, повторная синхронизация через rsync и пересборка gzip-архива

19. Сравнение содержимого архива, упакованного bzip2, с содержимым каталога /root/etc_backup:

```
root@eltex-practice2-pg2-v2:~# cd /root
root@eltex-practice2-pg2-v2:~# tar -djf etc_backup.tar.bz2 -C /root
etc_backup/motd: Mod time differs
etc_backup/motd: Size differs
root@eltex-practice2-pg2-v2:~# echo "tar --diff exit=$?"
tar --diff exit=1
root@eltex-practice2-pg2-v2:~# rm -rf /tmp/etc_bz2_cmp
root@eltex-practice2-pg2-v2:~# mkdir -p /tmp/etc_bz2_cmp
root@eltex-practice2-pg2-v2:~# tar -xjf etc_backup.tar.bz2 -C /tmp/etc_bz2_cm
p
root@eltex-practice2-pg2-v2:~# diff -r /tmp/etc_bz2_cmp/etc_backup /root/etc_
backup
diff -r /tmp/etc_bz2_cmp/etc_backup/motd /root/etc_backup/motd
2a3
> Обновлено 2026-10-25 20:27:50
diff: /tmp/etc_bz2_cmp/etc_backup/mtab: No such file or directory
diff: /root/etc_backup/mtab: No such file or directory
diff: /tmp/etc_bz2_cmp/etc_backup/os-release: No such file or directory
diff: /root/etc_backup/os-release: No such file or directory
diff: /tmp/etc_bz2_cmp/etc_backup/resolv.conf: No such file or directory
diff: /root/etc_backup/resolv.conf: No such file or directory
root@eltex-practice2-pg2-v2:~# echo "diff exit=$?"
diff exit=2
root@eltex-practice2-pg2-v2:~# diff /tmp/etc_bz2_cmp/etc_backup/motd /root/et
c_backup/motd
2a3
> Обновлено 2026-10-25 20:27:50
```

Рисунок 22 – Сравнение содержимого bzip2-архива с каталогом /root/etc_backup (tar --diff)

20. Распаковка архивов и сравнение файлов motd:

```
root@eltex-practice2-pg2-v2:~# ls -l /root/etc_backup_gzip/etc_backup/motd /r
oot/etc_backup_7zip/etc_backup/motd
-rw-r--r-- 1 root root 53 Oct 25 18:22 /root/etc_backup_7zip/etc_backup/motd
-rw-r--r-- 1 root root 92 Oct 25 20:27 /root/etc_backup_gzip/etc_backup/motd
root@eltex-practice2-pg2-v2:~# diff /root/etc_backup_gzip/etc_backup/motd /ro
ot/etc_backup_7zip/etc_backup/motd
3d2
< Обновлено 2026-10-25 20:27:50
root@eltex-practice2-pg2-v2:~# echo "diff exit=$?"
diff exit=1
```

Рисунок 23 – Распаковка gzip и 7z архивов, сравнение файлов motd с помощью diff (показано различие в датах)

Раздел 2

1. Поиск всех записей из лога загрузки в первые 20 секунд с момента загрузки:

```
root@eltex-practice2-pg2-v2:~# journalctl -b -o short-unix | awk 'NR==1 {star
t=$1} $1 <= start+20' | tee /tmp/boot20.txt | head -n 20
1792809896.102915 localhost kernel: Linux version 6.8.0-147-generic (buildd@l
cy02-amd64-081) (x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.
3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #147-Ubuntu SMP PREEMPT_DYNAMIC T
hu Sep 24 11:33:16 UTC 2026 (Ubuntu 6.8.0-147.147-generic 6.8.12)
1792809896.102947 localhost kernel: Command line: BOOT_IMAGE=/vmlinuz-6.8.0-1
47-generic root=/dev/mapper/ubuntu--vg-ubuntu--lv ro
1792809896.102964 localhost kernel: KERNEL supported cpus:
1792809896.102978 localhost kernel:   Intel GenuineIntel
1792809896.102993 localhost kernel:   AMD AuthenticAMD
1792809896.103007 localhost kernel:   Hygon HygonGenuine
1792809896.103023 localhost kernel:   Centaur CentaurHauls
1792809896.103037 localhost kernel:   zhaoxin   Shanghai
1792809896.103055 localhost kernel: BIOS-provided physical RAM map:
1792809896.103069 localhost kernel: BIOS-e820: [mem 0x0000000000000000-0x0000
00000009fbff] usable
1792809896.103083 localhost kernel: BIOS-e820: [mem 0x000000000009fc00-0x0000
00000009ffff] reserved
1792809896.103097 localhost kernel: BIOS-e820: [mem 0x00000000000f0000-0x0000
0000000fffff] reserved
1792809896.103111 localhost kernel: BIOS-e820: [mem 0x0000000000100000-0x0000
0000bffd9fff] usable
1792809896.103125 localhost kernel: BIOS-e820: [mem 0x00000000bffda000-0x0000
0000bfffffff] reserved
1792809896.103140 localhost kernel: BIOS-e820: [mem 0x00000000feffc000-0x0000
0000feffffff] reserved
1792809896.103154 localhost kernel: BIOS-e820: [mem 0x00000000fffc0000-0x0000
0000ffffffff] reserved
1792809896.103168 localhost kernel: BIOS-e820: [mem 0x0000000100000000-0x0000
00013fffffff] usable
1792809896.103182 localhost kernel: BIOS-e820: [mem 0x000000fd00000000-0x0000
00ffffffffff] reserved
1792809896.103195 localhost kernel: NX (Execute Disable) protection: active
1792809896.103209 localhost kernel: APIC: Static calls initialized
root@eltex-practice2-pg2-v2:~# wc -l /tmp/boot20.txt
43 /tmp/boot20.txt
```

Рисунок 24 – Поиск записей из лога загрузки (journalctl -b) за первые 20 секунд с помощью awk

2. Поиск всех источников и их сообщений в файле auth.log:

```
root@eltex-practice2-pg2-v2:~# find /var/log -name "auth.log*" 2>/dev/null
/var/log/auth.log
/var/log/auth.log.2.gz
/var/log/auth.log.1
root@eltex-practice2-pg2-v2:~# awk '{print $5, substr($0, index($0,$6))}' /va
r/log/auth.log | sed 's/\[[0-9]*\]//' | sort -k1,1 > /tmp/auth_sources.txt
root@eltex-practice2-pg2-v2:~# head -n 15 /tmp/auth_sources.txt
: command not allowed ; TTY=pts/2 ; PWD=/home/user1 ; USER=root ; COMMAND=dme
g
: TTY=pts/2 ; PWD=/home/user1 ; USER=root ; COMMAND=/usr/bin/dmesg
262 logged out. Waiting for processes to exit.
264 logged out. Waiting for processes to exit.
268 logged out. Waiting for processes to exit.
275 logged out. Waiting for processes to exit.
290 logged out. Waiting for processes to exit.
293 logged out. Waiting for processes to exit.
298 logged out. Waiting for processes to exit.
310 logged out. Waiting for processes to exit.
313 logged out. Waiting for processes to exit.
432 logged out. Waiting for processes to exit.
443 logged out. Waiting for processes to exit.
added to /etc/group: name=sharegrp, GID=1004
added to /etc/group: name=user2, GID=1002
```

Рисунок 25 – Поиск файла auth.log, извлечение источников и сообщений с помощью awk и sed, удаление PID

3. Поиск для результата предыдущего пункта количества повторений для каждого источника и вывод их в виде списка:

```
root@eltex-practice2-pg2-v2:~# awk '{print $1}' /tmp/auth_sources.txt | sort
| uniq -c | sort -nr > /tmp/auth_counts.txt
root@eltex-practice2-pg2-v2:~# head -n 15 /tmp/auth_counts.txt
    639 session
     12 publickey
     11 from
     11 disconnect
      7 user1)
      7 password
      6 added
      4 user3)
      4 group:
      3 user2)
      3 user:
      3 user
      2 'user3'
      2 'user2'
      2 'user1'
```

Рисунок 26 – Подсчёт количества повторений для каждого источника (sort | uniq -c | sort -nr)

4. Поиск в файле /etc/passwd всех пользователей в системе, у которых установлен shell /usr/sbin/nologin:

```
root@eltex-practice2-pg2-v2:~# awk -F: '$7 == "/usr/sbin/nologin" {print $3,
$1, $4}' /etc/passwd | sort -nr > /tmp/nologin_users.txt
root@eltex-practice2-pg2-v2:~# cat /tmp/nologin_users.txt
65534 nobody 65534
1003 user3 1003
998 systemd-network 998
997 systemd-timesync 997
992 systemd-resolve 992
991 polkitd 991
989 fwupd-refresh 989
109 sshd 65534
108 usbmux 46
107 landscape 109
105 tcpdump 107
104 uuidd 105
103 syslog 104
101 messagebus 102
42 _apt 65534
39 irc 39
38 list 38
34 backup 34
33 www-data 33
13 proxy 13
10 uucp 10
9 news 9
8 mail 8
7 lp 7
6 man 12
5 games 60
3 sys 3
2 bin 2
1 daemon 1
```

Рисунок 27 – Поиск пользователей с оболочкой /usr/sbin/nologin в /etc/passwd

```
root@eltex-practice2-pg2-v2:~# awk -F: 'NR==FNR {if ($7 == "/usr/sbin/nologin
") {user[$1]=$1; uid[$1]=$3; gid[$1]=$4}; next} {for (u in user) {if ($3 == g
id[u]) groups[u]=groups[u] $1 " "; split($4, a, ","); for (i in a) if (a[i] =
= u) groups[u]=groups[u] $1 " "}} END {for (u in user) {gsub(/ +/, " ", group
s[u]); print uid[u] ", " u ", " groups[u]}}' /etc/passwd /etc/group | sort -t
, -k1,1nr
65534, nobody, nogroup
1003, user3, users user3 sharegrp
998, systemd-network, systemd-network
997, systemd-timesync, systemd-timesync
992, systemd-resolve, systemd-resolve
991, polkitd, polkitd
989, fwupd-refresh, fwupd-refresh
109, sshd, nogroup
108, usbmux, plugdev
107, landscape, landscape
105, tcpdump, tcpdump
104, uuidd, uuidd
103, syslog, adm syslog
101, messagebus, messagebus
42, _apt, nogroup
39, irc, irc
38, list, list
34, backup, backup
33, www-data, www-data
13, proxy, proxy
10, uucp, uucp
9, news, news
8, mail, mail
7, lp, lp
6, man, man
5, games, games
3, sys, sys
2, bin, bin
1, daemon, daemon
```

Рисунок 28 – Формирование списка «UID, username, группы» для nologin-пользователей с сортировкой по убыванию UID

5. Поиск в результате вывода dmesg всех строк, содержащих слово «kernel»:

```
root@eltex-practice2-pg2-v2:~# dmesg | grep 'kernel' | head -n 10
[    0.118364] Booting paravirtualized kernel on KVM
[    0.242283] Memory: 3925848K/4193760K available (22528K kernel code, 4441K
 rwdata, 14440K rodata, 4936K init, 4772K bss, 267652K reserved, 0K cma-reser
ved)
[    0.953906] Loaded X.509 cert 'Build time autogenerated kernel key: 67dae3
bf0284ce2ad34b9632cb7c8734f4f57c6c'
[    1.273570] Loaded X.509 cert 'Build time autogenerated kernel key: 67dae3
bf0284ce2ad34b9632cb7c8734f4f57c6c'
[    1.315853] Freeing unused kernel image (initmem) memory: 4936K
[    1.317577] Write protecting the kernel read-only data: 38912k
[    1.319334] Freeing unused kernel image (rodata/data gap) memory: 1944K
[  106.023012] systemd[1]: Listening on systemd-udevd-kernel.socket - udev Ke
rnel Socket.
[  106.044443] systemd[1]: Mounting sys-kernel-debug.mount - Kernel Debug Fil
e System...
[  106.046466] systemd[1]: Mounting sys-kernel-tracing.mount - Kernel Trace F
ile System...
root@eltex-practice2-pg2-v2:~# wc -l /var/log/kern.log
0 /var/log/kern.log
```

Рисунок 29 – Поиск строк, содержащих слово 'kernel', в выводе dmesg

6. Подсчёт количества строк в файле /var/log/kern.log:

```
root@eltex-practice2-pg2-v2:~# wc -l /var/log/kern.log
0 /var/log/kern.log
```

Рисунок 30 – Подсчёт количества строк в файле /var/log/kern.log (файл пуст)

7. Форматирование вывода записей в /var/log/apt/history.log:

```
root@eltex-practice2-pg2-v2:~# awk -v RS='' -F'\n' '{sd=""; cmd=""; ed=""; fo
r(i=1;i<=NF;i++){if($i ~ /^Start-Date:/) sd=$i; if($i ~ /^Commandline:/) cmd=
$i; if($i ~ /^End-Date:/) ed=$i} if(cmd!="") print cmd " ; " sd " ; " ed}' /v
ar/log/apt/history.log | head -n 5
Commandline: apt-get install --reinstall apt ; Start-Date: 2025-02-16  20:51:
22 ; End-Date: 2025-02-16  20:51:24
Commandline: apt-get install --reinstall base-files ; Start-Date: 2025-02-16
 20:51:26 ; End-Date: 2025-02-16  20:51:28
Commandline: apt-get install --reinstall base-passwd ; Start-Date: 2025-02-16
  20:51:31 ; End-Date: 2025-02-16  20:51:32
Commandline: apt-get install --reinstall bash ; Start-Date: 2025-02-16  20:51
:35 ; End-Date: 2025-02-16  20:51:37
Commandline: apt-get install --reinstall bsdutils ; Start-Date: 2025-02-16  2
0:51:39 ; End-Date: 2025-02-16  20:51:41
```

Рисунок 31 – Форматирование вывода записей из /var/log/apt/history.log через awk

Раздел 3

2. Установка параметров git в соответствии с учётными данными:

```
root@eltex-practice2-pg2-v2:~# git config --global user.name "penakelex"
root@eltex-practice2-pg2-v2:~# git config --global user.email "alexeykoninsky
129@gmail.com"
root@eltex-practice2-pg2-v2:~# git config --global init.defaultBranch main
root@eltex-practice2-pg2-v2:~# git config --global --list
user.name=penakelex
user.email=alexeykoninsky129@gmail.com
init.defaultbranch=main
```

Рисунок 32 – Настройка глобальных параметров git (user.name, user.email, init.defaultBranch)

3. Установка git и gh:

```
root@eltex-practice2-pg2-v2:~# apt install -y git gh
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.4).
git set to manually installed.
The following NEW packages will be installed:
  gh
0 upgraded, 1 newly installed, 0 to remove and 94 not upgraded.
Need to get 8,836 kB of archives.
After this operation, 45.4 MB of additional disk space will be used.
Get:1 http://ru.archive.ubuntu.com/ubuntu noble-updates/universe amd64 gh amd
64 2.45.0-1ubuntu0.3 [8,836 kB]
Fetched 8,836 kB in 31s (288 kB/s)
Selecting previously unselected package gh.
(Reading database ... 125714 files and directories currently installed.)
Preparing to unpack .../gh_2.45.0-1ubuntu0.3_amd64.deb ...
Unpacking gh (2.45.0-1ubuntu0.3) ...
Setting up gh (2.45.0-1ubuntu0.3) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@eltex-practice2-pg2-v2:~# git --version
git version 2.43.0
root@eltex-practice2-pg2-v2:~# gh --version
gh version 2.45.0 (2025-07-18 Ubuntu 2.45.0-1ubuntu0.3)
https://github.com/cli/cli/releases/tag/v2.45.0
```

Рисунок 33 – Установка git и gh, проверка их версий

4. Создание ключа ssh:

```
root@eltex-practice2-pg2-v2:~# ssh-keygen -t ed25519 -f ~/.ssh/penakelex
Generating public/private ed25519 key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/penakelex
Your public key has been saved in /root/.ssh/penakelex.pub
The key fingerprint is:
SHA256:xdFrTALuxr7IQCvCbfDChmKBmbU6jswk8jYQMlKRzL4 root@eltex-practice2-pg2-v
2
The key's randomart image is:
+--[ED25519 256]--+
| o.o    ....     |
|  =    . ...o    |
| o.     . o+ .   |
|*+..   o .  +    |
|*=...   S  .     |
|=.E. . o         |
|OO.+o   .        |
|%+=. o . .       |
|.=..  o .        |
+----[SHA256]-----+
root@eltex-practice2-pg2-v2:~# ls -l ~/.ssh/penakelex*
-rw------- 1 root root 419 Oct 26 15:24 /root/.ssh/penakelex
-rw-r--r-- 1 root root 109 Oct 26 15:24 /root/.ssh/penakelex.pub
root@eltex-practice2-pg2-v2:~# cat ~/.ssh/penakelex.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPHI0pimt5PaKGZuDrrcYU7rkSd2HU46xWc+aBgKJ
8dV root@eltex-practice2-pg2-v2
```

Рисунок 34 – Создание ключа SSH ed25519 специально для GitHub и проверка его наличия

5. Подключение компьютера к GitHub:

```
root@eltex-practice2-pg2-v2:~# gh auth login
? What account do you want to log into? GitHub.com
? What is your preferred protocol for Git operations on this host? SSH
? Upload your SSH public key to your GitHub account? /root/.ssh/penakelex.pub
? Title for your SSH key: eltex-practice
? How would you like to authenticate GitHub CLI? Login with a web browser

! First copy your one-time code: 21FB-B2CD
Press Enter to open github.com in your browser...
! Failed opening a web browser at https://github.com/login/device
  exec: "xdg-open,x-www-browser,www-browser,wslview": executable file not fou
nd in $PATH
  Please try entering the URL in your browser manually
✓ Authentication complete.
- gh config set -h github.com git_protocol ssh
✓ Configured git protocol
! Authentication credentials saved in plain text
✓ Uploaded the SSH key to your GitHub account: /root/.ssh/penakelex.pub
✓ Logged in as penakelex
root@eltex-practice2-pg2-v2:~# gh auth status
github.com
  ✓ Logged in to github.com account penakelex (/root/.config/gh/hosts.yml)
  - Active account: true
  - Git operations protocol: ssh
  - Token: gho_************************************
  - Token scopes: 'admin:public_key', 'gist', 'read:org', 'repo'
```

Рисунок 35 – Авторизация в GitHub CLI через gh auth login (выбор SSH, загрузка ключа и вход через браузер)

6. Автоматическая авторизация для github.com:

```
root@eltex-practice2-pg2-v2:~# gh auth setup-git --hostname github.com
```

Рисунок 36 – Установка автоматической авторизации для git через gh auth setup-git

7. Проверка статуса:

```
root@eltex-practice2-pg2-v2:~# gh auth status
github.com
  ✓ Logged in to github.com account penakelex (/root/.config/gh/hosts.yml)
  - Active account: true
  - Git operations protocol: ssh
  - Token: gho_************************************
  - Token scopes: 'admin:public_key', 'gist', 'read:org', 'repo'
```

Рисунок 37 – Проверка статуса авторизации GitHub CLI (gh auth status)

8. Создание репозитория на сервере GitHub:

```
root@eltex-practice2-pg2-v2:~# gh repo create eltex-module1-practice2 --publi
c
✓ Created repository penakelex/eltex-module1-practice2 on GitHub
  https://github.com/penakelex/eltex-module1-practice2
```

Рисунок 38 – Создание публичного репозитория eltex-module1-practice2 на GitHub

9. Наполнение репозитория:

```
root@eltex-practice2-pg2-v2:~# mkdir ~/eltex-module1-practice2 && cd ~/eltex-
module1-practice2
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# echo "# eltex-module1-
practice2" >> README.md
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git init
Initialized empty Git repository in /root/eltex-module1-practice2/.git/
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git add README.md
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git commit -m "first c
ommit"
[main (root-commit) d7252dc] first commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git branch -M main
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git remote add origin
https://github.com/penakelex/eltex-module1-practice2.git
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git status
On branch main
nothing to commit, working tree clean
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git push -u origin mai
n
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 244 bytes | 244.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/penakelex/eltex-module1-practice2.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# mkdir -p module1/task1
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# mkdir module1/task2
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# mkdir module1/task3
```

Рисунок 39 – Создание локального репозитория, первый коммит и отправка на GitHub (git init, git push)

```
eltex-pg2-v2@eltex-2026-autumn:~$ scp ~/report_part1.txt root@172.16.9.202:~/
eltex-module1-practice2/module1/task1/
report_part1.txt                              100%  132     5.1KB/s   00:00
eltex-pg2-v2@eltex-2026-autumn:~$ scp ~/eltex-pg2-v2_part.log root@172.16.9.2
02:~/eltex-module1-practice2/module1/task2/
eltex-pg2-v2_part.log                         100%  788     2.3MB/s   00:00
eltex-pg2-v2@eltex-2026-autumn:~$ scp ~/eltex-pg2-v2_part3.log root@172.16.9.
202:~/eltex-module1-practice2/module1/task3/
eltex-pg2-v2_part3.log                        100%  305   380.3KB/s   00:00
eltex-pg2-v2@eltex-2026-autumn:~$ scp ~/eltex-pg2-v2_part3_gt20M.log root@172
.16.9.202:~/eltex-module1-practice2/module1/task3/
eltex-pg2-v2_part3_gt20M.log                  100% 1129    60.0KB/s   00:00
```

Рисунок 40 – Копирование отчетов по первой практике с сервера 1 на сервер 2 с помощью scp

```
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# ls -l module1/task*
module1/task1:
total 4
-rw-r--r-- 1 root root 132 Oct 26 16:25 report_part1.txt

module1/task2:
total 4
-rw-r--r-- 1 root root 788 Oct 26 16:25 eltex-pg2-v2_part.log

module1/task3:
total 8
-rw-r--r-- 1 root root 1129 Oct 26 16:25 eltex-pg2-v2_part3_gt20M.log
-rw-r--r-- 1 root root  305 Oct 26 16:25 eltex-pg2-v2_part3.log
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git add .
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   module1/task1/report_part1.txt
        new file:   module1/task2/eltex-pg2-v2_part.log
        new file:   module1/task3/eltex-pg2-v2_part3.log
        new file:   module1/task3/eltex-pg2-v2_part3_gt20M.log

root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git commit -m "Added r
eports for module 1"
[main 23a7464] Added reports for module 1
 4 files changed, 104 insertions(+)
 create mode 100644 module1/task1/report_part1.txt
 create mode 100644 module1/task2/eltex-pg2-v2_part.log
 create mode 100644 module1/task3/eltex-pg2-v2_part3.log
 create mode 100644 module1/task3/eltex-pg2-v2_part3_gt20M.log
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git push
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 2 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (10/10), 1.61 KiB | 1.61 MiB/s, done.
Total 10 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/penakelex/eltex-module1-practice2.git
   d7252dc..23a7464  main -> main
```

Рисунок 41 – Добавление отчетов в репозиторий (git add, git commit) и отправка на GitHub

10. Клонирование репозитория:

```
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# cd ~
root@eltex-practice2-pg2-v2:~# mkdir -p eltex-module1-practice2_clone
root@eltex-practice2-pg2-v2:~# cd eltex-module1-practice2_clone
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2_clone# git clone https:
//github.com/penakelex/eltex-module1-practice2.git .
Cloning into '.'...
remote: Enumerating objects: 13, done.
remote: Counting objects: 100% (13/13), done.
remote: Compressing objects: 100% (9/9), done.
remote: Total 13 (delta 0), reused 13 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (13/13), done.
```

Рисунок 42 – Клонирование репозитория в отдельный каталог

11. Внесение изменения в основном каталоге и выполнение fetch, log, diff в клоне:

```
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2_clone# cd ~/eltex-modul
e1-practice2
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# echo "Это изменение вн
есено из основного каталога" >> README.md
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git add README.md
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git commit -m "Update
README from main directory"
[main e876b16] Update README from main directory
 1 file changed, 1 insertion(+)
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 392 bytes | 392.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/penakelex/eltex-module1-practice2.git
   23a7464..e876b16  main -> main
```

Рисунок 43 – Внесение изменений в основной каталог, коммит и git push

```
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# cd ~/eltex-module1-pra
ctice2_clone
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2_clone# git fetch
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 372 bytes | 372.00 KiB/s, done.
From https://github.com/penakelex/eltex-module1-practice2
   23a7464..e876b16  main       -> origin/main
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2_clone# git log origin/m
ain
commit e876b1657fd84bac380a4f24b03cdfcff5c787e4 (origin/main, origin/HEAD)
Author: penakelex <alexeykoninsky129@gmail.com>
Date:   Mon Oct 26 16:30:18 2026 +0000

    Update README from main directory

commit 23a7464569dd1d8e93cffa3aa5158e31ff30a3e4 (HEAD -> main)
Author: penakelex <alexeykoninsky129@gmail.com>
Date:   Mon Oct 26 16:26:19 2026 +0000

    Added reports for module 1

commit d7252dcc74741578f98e36b1bcf1348c600d1b4c
Author: penakelex <alexeykoninsky129@gmail.com>
Date:   Mon Oct 26 15:58:18 2026 +0000

    first commit
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2_clone# git diff main or
igin/main
diff --git a/README.md b/README.md
index 7d2cfdb..93c7247 100644
--- a/README.md
+++ b/README.md
@@ -1 +1,2 @@
 # eltex-module1-practice2
+Это изменение внесено из основного каталога
```

Рисунок 44 – Получение обновлений в клоне (git fetch), просмотр логов и разницы (git log, git diff)

```
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2_clone# git merge origin
/main
Updating 23a7464..e876b16
Fast-forward
 README.md | 1 +
 1 file changed, 1 insertion(+)
```

Рисунок 45 – Слияние изменений в клоне (git merge)

12. Заполнение репозитория отчётами по второй практической работе:

```
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# mkdir -p module1/pract
ice2/task1
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# mkdir -p module1/pract
ice2/task2
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# mkdir -p module1/pract
ice2/task3
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# cp /root/practice2_elt
ex-pg2-v2_part1.log module1/practice2/task1/
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# cp /root/practice2_elt
ex-pg2-v2_part2.log module1/practice2/task2/
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# cp /root/practice2_elt
ex-pg2-v2_part3.log module1/practice2/task3/
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git add .
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   module1/practice2/task1/practice2_eltex-pg2-v2_part1.log
        new file:   module1/practice2/task2/practice2_eltex-pg2-v2_part2.log
        new file:   module1/practice2/task3/practice2_eltex-pg2-v2_part3.log

root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git commit -m "Added p
ractice 2 reports"
[main 7eabe3a] Added practice 2 reports
 3 files changed, 488 insertions(+)
 create mode 100644 module1/practice2/task1/practice2_eltex-pg2-v2_part1.log
 create mode 100644 module1/practice2/task2/practice2_eltex-pg2-v2_part2.log
 create mode 100644 module1/practice2/task3/practice2_eltex-pg2-v2_part3.log
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# git push
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 2 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (10/10), 9.21 KiB | 9.21 MiB/s, done.
Total 10 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/penakelex/eltex-module1-practice2.git
   e876b16..7eabe3a  main -> main
```

Рисунок 46 – Создание каталогов для отчётов второй практики, копирование логов, коммит и git push

13. Изначально репозиторий создавался под конкретную практику (eltex-module1-practice2), однако для создания единого пространства для всех последующих модулей было решено использовать общее имя. Старый репозиторий eltex-practices удалён, текущий репозиторий переименован в eltex-practices, локальная папка также переименована. Ссылка, отправленная преподавателю ранее, автоматически перенаправляет на обновлённый репозиторий. Все отчёты (и первой, и второй практики) сохранены в этом репозитории.

```
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# gh repo delete penakel
ex/eltex-practices --yes
✓ Deleted repository penakelex/eltex-practices
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# gh repo rename eltex-p
ractices --yes
✓ Renamed repository penakelex/eltex-practices
✓ Updated the "origin" remote
root@eltex-practice2-pg2-v2:~/eltex-module1-practice2# cd ~
root@eltex-practice2-pg2-v2:~# mv ~/eltex-module1-practice2 ~/eltex-practices
root@eltex-practice2-pg2-v2:~# cd ~/eltex-practices
root@eltex-practice2-pg2-v2:~/eltex-practices# git remote -v
origin  git@github.com:penakelex/eltex-practices.git (fetch)
origin  git@github.com:penakelex/eltex-practices.git (push)
root@eltex-practice2-pg2-v2:~/eltex-practices# git push
The authenticity of host 'github.com (140.82.121.4)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU
.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?yes^[[D^[
^[[D^[[A
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
git@github.com: Permission denied (publickey).
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
root@eltex-practice2-pg2-v2:~/eltex-practices# ls ~/.ssh
authorized_keys  known_hosts  penakelex  penakelex.pub
root@eltex-practice2-pg2-v2:~/eltex-practices# echo -e "Host github.com\n  Id
entityFile ~/.ssh/penakelex\n  IdentitiesOnly yes" >> ~/.ssh/config
root@eltex-practice2-pg2-v2:~/eltex-practices# chmod 600 ~/.ssh/config
root@eltex-practice2-pg2-v2:~/eltex-practices# ssh -T git@github.com
Hi penakelex! You've successfully authenticated, but GitHub does not provide
shell access.
root@eltex-practice2-pg2-v2:~/eltex-practices# git push
Everything up-to-date
```

Рисунок 47 – Удаление старого репозитория, переименование текущего в eltex-practices, настройка SSH-конфига и финальный git push

14. При выполнении п. 12 (создание каталогов для отчётов 2-й практики) было принято решение вынести их в отдельный подкаталог module1/practice2/, чтобы избежать конфликта имён с каталогами task1-3, созданными ранее для 1-й практики. Для поддержания единой структуры было решено привести каталоги к единому виду: отчёты первой практики перенесены в module1/practice1/. Выполнены команды перемещения каталогов, изменения зафиксированы в git и отправлены на GitHub.

```
root@eltex-practice2-pg2-v2:~/eltex-practices# mkdir -p module1/practice1
root@eltex-practice2-pg2-v2:~/eltex-practices# mv module1/task1 module1/task2
 module1/task3 module1/practice1/
root@eltex-practice2-pg2-v2:~/eltex-practices# git add .
root@eltex-practice2-pg2-v2:~/eltex-practices# git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        renamed:    module1/task1/report_part1.txt -> module1/practice1/task1
/report_part1.txt
        renamed:    module1/task2/eltex-pg2-v2_part.log -> module1/practice1/
task2/eltex-pg2-v2_part.log
        renamed:    module1/task3/eltex-pg2-v2_part3.log -> module1/practice1
/task3/eltex-pg2-v2_part3.log
        renamed:    module1/task3/eltex-pg2-v2_part3_gt20M.log -> module1/pra
ctice1/task3/eltex-pg2-v2_part3_gt20M.log

root@eltex-practice2-pg2-v2:~/eltex-practices# git commit -m "Organized direc
tories: moved practice 1 reports to module1/practice1"
[main 56cf66f] Organized directories: moved practice 1 reports to module1/pra
ctice1
 4 files changed, 0 insertions(+), 0 deletions(-)
 rename module1/{ => practice1}/task1/report_part1.txt (100%)
 rename module1/{ => practice1}/task2/eltex-pg2-v2_part.log (100%)
 rename module1/{ => practice1}/task3/eltex-pg2-v2_part3.log (100%)
 rename module1/{ => practice1}/task3/eltex-pg2-v2_part3_gt20M.log (100%)
root@eltex-practice2-pg2-v2:~/eltex-practices# git push
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 475 bytes | 475.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:penakelex/eltex-practices.git
   7eabe3a..56cf66f  main -> main
```

Рисунок 48 – Унификация структуры каталогов в репозитории